



Kuwait University

Faculty of Science

Department of Mathematics

Math 102: Calculus II

Fall 2021–2022

Second Mid-Term Exam
Saturday, January 08, 2022

Name										
Student ID										
Section No.						Serial No.				

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- The duration of the exam is **90 minutes**.
 - Give full justification for your answers and state clearly any theorem you use.
 - Calculators and communication devices are not allowed in the examination room.
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Question 1 (15 pts)	
Question 2 (15 pts)	
Question 3 (15 pts)	
Question 4 (15 pts)	
Question 5 (15 pts)	
Question 6 (15 pts)	
Question 7 (10 pts)	
Total	100

1. [15 pts.] Determine whether the series

$$\sum_{n=1}^{\infty} \left(\frac{2}{n(n+1)} + \frac{e^n}{3^{n-1}} \right)$$

is convergent or divergent. If it is convergent, find its sum.

2. [15 pts.] Use the integral test to determine whether the series $\sum_{n=2}^{\infty} \frac{1}{n(\ln n)^2}$ is convergent or divergent.

3. [15 pts.] Determine whether the series $\sum_{n=1}^{\infty} (-1)^{n-1} \frac{1}{\sqrt{5n+1}}$ is absolutely convergent, conditionally convergent, or divergent.

4. [15 pts.] Find the radius of convergence and interval of convergence of the series

$$\sum_{n=1}^{\infty} (-1)^{n-1} \frac{(x-3)^n}{2n+1}.$$

5. [10 + 5 = 15 pts.] (a) Use differentiation to find a power series representation for $f(x) = \frac{1}{(1+x)^2}$ and determine the radius of convergence.

- (b) Use part (a) to find the sum of $\sum_{n=1}^{\infty} (-1)^n \frac{n}{3^{n-1}}$.

6. [6 + 9 = 15 pts.] A sequence $\{a_n\}$ is given by $a_1 = 1$, $a_{n+1} = \frac{a_n}{2 + a_n}$, $n = 1, 2, \dots$.

(a) Show that $\{a_n\}$ is decreasing and bounded below.

(b) Apply the Monotonic Sequence Theorem to show that $\lim_{n \rightarrow \infty} a_n$ exists and find its limit.

7. [10 pts.] Show that if the series $\sum_{n=1}^{\infty} a_n$ is convergent, then $\lim_{n \rightarrow \infty} a_n = 0$.